An upper bound on the mean value of the Erdős-Hooley delta function

Terence Tao · 16 June 2023

https://terrytao.wordpress.com/2023/06/16/a-note-on-the-mean-value-of-the-hooley-delta-function/

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Dimitris Koukoulopoulos and I have just uploaded to the arXiv our paper "An upper bound on the mean value of the Erdős-Hooley delta function". This paper concerns a (still somewhat poorly understood) basic arithmetic function in multiplicative number theory, namely the *Erdos-Hooley delta function*

$$\Delta(n) := \sup_u \Delta(n;u)$$

where

$$\Delta(n;u) := \#\{d \mid n : e^u < d \leq e^{u+1}\}.$$

The function Δ measures the extent to which the divisors of a natural number can be concentrated in a dyadic (or more precisely, e-dyadic) interval $(e^u, e^{u+1}]$. From the pigeonhole principle, we have the bounds

$$\frac{\tau(n)}{\log n} \ll \Delta(n) \leq \tau(n),$$

where $\tau(n) := \#\{d : d \mid n\}$ is the usual divisor function. The statistical behavior of the divisor function is well understood; for instance, if n is drawn at random from 1 to x, then the mean value of $\tau(n)$ is roughly $\log x$, the median is roughly $\log^{\log 2} x$, and (by the Erdős-Kac theorem) $\tau(n)$ asymptotically has a log-normal distribution. In particular, there are a small proportion of highly divisible numbers that skew the mean to be significantly higher than the median.

On the other hand, the statistical behavior of the Erdős-Hooley delta function is significantly less well understood, even conjecturally. Again drawing n at random from 1 to x for large x, the median is known to be somewhere between $(\log\log x)^{0.3533\ldots}$ and $(\log\log x)^{0.6102\ldots}$ for large x – a (difficult) recent result of Ford, Green, and Koukoulopolous (for the lower bound) and La Bretèche and Tenenbaum (for the upper bound). And the mean $\frac{1}{x}\sum_{n\leq x}\Delta(n)$ was even less well controlled; the best previous bounds were

$$\log\log x \ll \frac{1}{x}\sum_{n\leq x}\Delta(n) \ll \exp(c\sqrt{\log\log x})$$

for any $c > \sqrt{2}\log 2$, with the lower bound due to Hall and Tenenbaum, and the upper bound a recent result of La Bretèche and Tenenbaum.

The main result of this paper is an improvement of the upper bound to

$$\frac{1}{x}\sum_{n\leq x}\Delta(n) \ll (\log\log x)^{11/4}.$$

It is still unclear to us exactly what to conjecture regarding the actual order of the mean value.

The reason we looked into this problem was that it was connected to forthcoming work of David Conlon, Jacob Fox, and Huy Pham on the following problem of Erdos: what is the size of the largest subset A of $\{1,\dots,N\}$ with the property that no non-empty subset of A sums to a perfect square? Erdos observed that one can obtain sets of size $\gg N^{1/3}$ (basically by considering certain homogeneous arithmetic progressions), and Nguyen and Vu showed an upper bound of $\ll N^{1/3}(\log N)^{O(1)}$. With our mean value bound as input, together with several new arguments, Conlon, Fox, and Pham have been able to improve the upper bound to $\ll N^{1/3}(\log\log N)^{O(1)}$).

Let me now discuss some of the ingredients of the proof. The first few steps are standard. Firstly we may restrict attention to square-free numbers without much difficulty (the point being that if a number n factors as $n = d^2m$ with m squarefree, then $\Delta(n) \leq \tau(d^2)\Delta(m)$). Next, because a square-free number $n > 1$ can be uniquely factored as $n = pm$ where p is a prime and m lies in the finite set $\mathcal{S}_{<p}$ of squarefree numbers whose prime factors are less than p, and $\Delta(n) \leq \tau(p)\Delta(m) = 2\Delta(m)$, it is not difficult to establish the bound

$$\frac{1}{x}\sum_{n\in\mathcal{S}_{<x}}\Delta(n) \ll \sup_{2\leq y\leq x}\frac{1}{\log y}\sum_{n\in\mathcal{S}_{<y}}\frac{\Delta(n)}{n}.$$

The upshot of this is that one can replace an ordinary average with a logarithmic average, thus it suffices to show

$$\frac{1}{\log x}\sum_{n\in\mathcal{S}_{<x}}\frac{\Delta(n)}{n} \ll (\log\log x)^{11/4}. \tag{1}$$

We actually prove a slightly more refined distributional estimate: for any $A \geq 2$, we have a bound

$$\Delta(n) \ll A\log^{3/4} A \tag{2}$$

outside of an exceptional set E which is small in the sense that

$$\frac{1}{\log x}\sum_{n\in\mathcal{S}_{<x}:n\in E}\frac{1}{n} \ll \frac{1}{A}. \tag{3}$$

It is not difficult to get from this distributional estimate to the logarithmic average estimate (1) (worsening the exponent $3/4$ to $3/4+2 = 11/4$).

To get some intuition on the size of $\Delta(n)$, we observe that if $y > 0$ and $n_{<y}$ is the factor of n coming from the prime factors less than y, then

$$\Delta(n) \geq \Delta(n_{<y}) \gg \frac{\tau(n_{<y})}{\log n_{<y}}. \tag{4}$$

On the other hand, standard estimates let one establish that

$$\tau(n_{<y}) \ll A \log n_{<y} \tag{5}$$

for all y, and all n outside of an exceptional set that is small in the sense (3); in fact it turns out that one can also get an additional gain in this estimate unless $\log y$ is close to $A^{\log 4}$, which turns out to be useful when optimizing the bounds. So we would like to approximately reverse the inequalities in (4) and get from (5) to (2), possibly after throwing away further exceptional sets of size (3).

At this point we perform another standard technique, namely the moment method of controlling the supremum $\Delta(n) = \sup_u \Delta(n;u)$ by the moments

$$M_q(n) := \int_{\mathbf{R}} \Delta(n,u)^q \, du$$

for natural numbers q; it is not difficult to establish the bound

$$\Delta(n) \ll M_q(n)^{1/q}$$

and one expects this bound to become essentially sharp once $q \sim \log\log x$. We will be able to show a moment bound

$$\sum_{n \in \mathcal{S}_{<x} \backslash E_q} \frac{M_q(n)/\tau(n)}{n} \leq O(q)^q A^{q-2} \log^{3q/4} A$$

for any $q \geq 2$ for some exceptional set E_q obeying the smallness condition (3) (actually, for technical reasons we need to improve the right-hand side slightly to close an induction on q); this will imply the distributional bound (2) from a standard Markov inequality argument (setting $q \sim \log\log x$).

The strategy is then to obtain a good recursive inequality for (averages of) $M_q(n)$. As in the reduction to (1), we factor $n = pm$ where p is a prime and $m \in \mathcal{S}_{<p}$. One observes the identity

$$\Delta(n;u) = \Delta(m;u) + \Delta(m;u-\log p)$$

for any u; taking moments, one obtains the identity

$$M_q(n) = \sum_{a+b=q; 0 \leq b \leq q} \binom{q}{a} \int_{\mathbf{R}} \Delta(m;u)^a \Delta(m;u-\log p)^b \, du.$$

As in previous literature, one can try to average in p here and apply Hölder's inequality. But it convenient to first use the symmetry of the summand in a,b to reduce to the case of relatively small values of b:

$$M_q(n) \leq 2 \sum_{a+b=q; 0 \leq b \leq q/2} \binom{q}{a} \int_{\mathbf{R}} \Delta(m;u)^a \Delta(m;u-\log p)^b \, du.$$

One can extract out the $b=0$ term as

$$M_q(n) \leq 2M_q(m)$$
$$+2 \sum_{a+b=q; 1 \leq b \leq q/2} \binom{q}{a} \int_{\mathbf{R}} \Delta(m;u)^a \Delta(m; u - \log p)^b \, du.$$

It is convenient to eliminate the factor of 2 by dividing out by the divisor function:

$$\frac{M_q(n)}{\tau(n)} \leq \frac{M_q(m)}{\tau(m)}$$
$$+\frac{1}{m} \sum_{a+b=q; 1 \leq b \leq q/2} \binom{q}{a} \int_{\mathbf{R}} \Delta(m;u)^a \Delta(m; u - \log p)^b \, du.$$

This inequality is suitable for iterating and also averaging in p and m. After some standard manipulations (using the Brun–Titchmarsh and Hölder inequalities), one is able to estimate sums such as

$$\sum_{n \in \mathcal{S}_{<x} \backslash E_q} \frac{M_q(n)/\tau(n)}{n} \tag{6}$$

in terms of sums such as

$$\int_2^{x^2} \sum_{a+b=q; 1 \leq b \leq q/2} \binom{q}{a} \sum_{n \in \mathcal{S}_{<x} \backslash E_q} \frac{M_a(n) M_b(n)}{\tau(n) n} \frac{dy}{\log^2 y}$$

(assuming a certain monotonicity property of the exceptional set E_q that turns out to hold in our application). By an induction hypothesis and a Markov inequality argument, one can get a reasonable pointwise upper bound on M_b (after removing another exceptional set), and the net result is that one can basically control the sum (6) in terms of expressions such as

$$\sum_{n \in \mathcal{S}_{<x} \backslash E_a} \frac{M_a(n)/\tau(n)}{n}$$

for various $a < q$. This allows one to estimate these expressions efficiently by induction.

Comments (13)



Ben Green · 16 June 2023

Terry,

Just to remark that the a.s. upper bound you state is in fact due to Tenenbaum and la Breteche https://arxiv.org/abs/2210.13897

Best wishes,
Ben

[Corrected, thanks – T.]

Anonymous · 16 June 2023

Is there any simple sufficient condition for a general class of functions $f(n)$ which enable estimating its mean by its logarithmic mean (as done for $Delta(n)$)?



Terence Tao · 17 June 2023

You can do this for any non-negative function f obeying a bound of the form $f(dm) \ll f(m)\tau(d)^{O(1)}$, by some standard elementary calculations:

$$\sum_{n\leq x} \log n f(n) = \sum_{dm\leq x} \Lambda(d) f(dm)$$
$$\ll \sum_{m\leq x} f(m) \sum_{d\leq x/m} \Lambda(d)\tau(d)^{O(1)}$$
$$\ll \sum_{m\leq x} f(m)\frac{x}{m}$$

and also

$$\sum_{n\leq x} \log \frac{x}{n} f(n) \ll \sum_{n\leq x} \frac{x}{n} f(n)$$

which upon summing gives

$$\frac{1}{x}\sum_{n\leq x} f(n) \ll \frac{1}{\log x} \sum_{n\leq x} \frac{f(n)}{n}.$$



David Roberts · 16 June 2023

Typo in the definition of Delta: l instead of ;

[Corrected, thanks – T.]

Lior Silberman · 17 June 2023

In the first displayed equation the l should be a semicolon.

[Corrected; thanks -T.]

Anonymous · 17 June 2023

You misspelled your coauthor's name…

[Corrected, thanks – T.]

Arman · 18 June 2023

PROF only IQ 200+ can understand what U wrote 🤗

Arman · 18 June 2023

PROF UR too smart i can't understand what U wrote

Arman · 18 June 2023

PROF why my comments getting delated😏

abc · 18 June 2023

the link to [Nguyen and Vu] does not work.

[Corrected, thanks – T.]



A lower bound on the mean value of the Erd”os-Hooley delta function | What's new · 23 August 2023

[…] bound on the mean value of the Erdös-Hooley delta function“. This paper complements the recent paper of Dimitris and myself obtaining the upper […]



Anonymous · 27 February 2026

Sorry if I am missing something obvious, but it seems to me that there is a typo in your equation (3), and I can’t figure out what you meant there. Did you just mean

$$\frac{1}{\log x} \sum_{n \in E} \frac{1}{n} \ll \frac{1}{A}$$

or do you also want n to be in $S_{<x}$? The set on which you are summing isn’t completely clear to me.



Terence Tao · 27 February 2026

The typo has now been corrected. One is summing over $E \cap S_{<x}$ where E is an exceptional set that does not depend on x.